

PHOEBE CHINYERU ACHONWA

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SUMMARY

M.Eng. student in Electrical Engineering at UCI (Dec 2026) with hands-on experience in analog/mixed-signal circuits, embedded systems, and hardware validation. Designed CMOS analog blocks including CDRs, differential amplifiers, and oscillators in 45nm technology using Cadence Virtuoso. Experienced in circuit simulation, performance analysis, automation scripting, and electromechanical system integration. Seeking hardware, analog, and validation engineering roles.

EDUCATION

University of California, Irvine - M. Eng. in Electrical Engineering

Irvine, CA

Concentration: Communications Circuits and Systems

Sep 2025 – Dec 2026

- **Coursework:** High Freq IC, Broadband IC, Adv. Analog IC Design, Adv. Semiconductor Physics, Micro-Sensors and Actuators (MEMS)

University of California, Santa Cruz - B.S. in Computer Engineering

Santa Cruz, CA

Concentration: Computer Systems

Sep 2018 – June 2022

PROJECTS

- **110GHz Push-Push Harmonic VCO & Buffer Chain:** Designed a push-push cross-coupled oscillator using transistor parasitics and transmission lines with a 3 stage cascode buffer amplifier. Achieved 110GHz second harmonic oscillation frequency, +7.3dBm output power, phase noise of -68 dBc/Hz and 10% tuning with MOS varactors.
- **10Gbps Clock and Data Recovery Circuit:** Designed a Type-II CDR with an LC-VCO, CMOS implementation of an Alexander phase detector and charge pump with a settling time $\leq 20\mu\text{s}$ and phase margin greater than 60° . Implemented in 45nm CMOS using Cadence Virtuoso.
- **MEMS Resonant Accelerometer Design:** Modeled a MEMS resonant accelerometer with a DETF resonator, cantilever suspension system, double-sided lateral comb drive actuator and proof mass operating at 10kHz. Layouts created with the polymumps fabrication rules in L-Edit and modal analyses in CoventorWare.
- **Two-Stage CMOS Amplifier:** Designed and simulated a differential telescopic amplifier and source follower buffer stage targeting GBW $> 350\text{ GHz}$, CMRR $> 80\text{ dB}$, phase margin $> 60^\circ$ and DC power $< 12\text{mW}$. Implemented in 45nm CMOS using Cadence Virtuoso/Spectre.
- **Mechatronics Capstone Project:** Developed embedded control architecture for an autonomous robot integrating motors, RC servos, optical sensors, and mechanical actuators. Implemented sensor interfacing using ADC thresholding, hysteresis, and debouncing. Debugged hardware/software integration issues by analyzing sensor behavior, control logic instability, and system-level performance.

WORK EXPERIENCE

UCI Clean Energy Institute - Master's Program Sponsored Project

Irvine, CA

Team Member

Jan 2026 – June 2026

- Automated a 3-axis system with a Raspberry Pi, Arduino, stepper motors, servos and custom control software for Proton Exchange Membrane Fuel Cells (PEMFC) catalyst deposition.
- Implemented distributed power architecture for power regulation and electrical isolation to prevent communication faults and undervoltage due to switching noise.
- Developed full-stack control application with Python/Tkinter for G-Code path generation, and real-time hardware monitoring with OctoPrint REST API.

Industrial Light and Magic - Technical Assistant

Feb 2023 – Sep 2025

- Developed Python automation and data collection tools for production pipelines, including a redesigned DRM data aggregator used by managers globally.

SKILLS

Hardware: CMOS analog circuit design, performance analyses, phase-locked loop (PLL), VCOs, CDRs, power management

Embedded Systems: Microcontrollers (PIC32, Arduino, Raspberry Pi), embedded Linux, hierarchical state machines

Software & Tools: Python, C (embedded), Cadence Virtuoso, L-Edit, CoventorWare